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12. Implement Iris Flower Classification using KNN

a) Iris Dataset 1

	Sepal Width	Petal Length	Petal Width	Species
Sepal Length				
5.1	3.5	1.4	0.2	Setosa
4.9	3.0	1.4	0.2	Setosa
6.0	2.9	4.5	1.5	Versicolor
5.9	3.0	4.2	1.5	Versicolor
6.5	3.0	5.2	2.0	Virginica
6.7	3.1	5.6	2.4	Virginica

PROGRAM

```
import pandas as pd
from sklearn.neighbors import KNeighborsClassifier
df = pd.read_csv("iris1.csv")
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
model = KNeighborsClassifier(n_neighbors=3)
model.fit(X, y)
sample = [[6.4, 3.0, 5.1, 1.9]]
print("Predicted Flower:", model.predict(sample)[0])
```

OUTPUT

Predicted Flower: Virginica

b) Iris Dataset 2

Sepal Length	Sepal Width	Petal Length	Petal Width	Species
5.0	3.4	1.5	0.2	Setosa
5.2	3.5	1.5	0.2	Setosa
5.8	2.7	4.1	1.0	Versicolor
6.1	2.8	4.0	1.3	Versicolor
6.3	2.9	5.6	1.8	Virginica
6.7	3.0	5.2	2.3	Virginica

PROGRAM

```
import pandas as pd
from sklearn.neighbors import KNeighborsClassifier
df = pd.read_csv("iris2.csv")
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
model = KNeighborsClassifier(n_neighbors=3)
model.fit(X, y)
sample = [[5.1, 3.4, 1.5, 0.2]]
print("Predicted Flower:", model.predict(sample)[0])
```

OUTPUT

Predicted Flower: Setosa

c) Iris Dataset 3

Sepal Length	Sepal Width	Petal Length	Petal Width	Species
5.4	3.7	1.5	0.2	Setosa
4.8	3.4	1.6	0.2	Setosa
6.0	2.7	5.1	1.6	Versicolor
5.7	2.8	4.5	1.3	Versicolor
6.2	3.4	5.4	2.3	Virginica
6.4	3.2	5.3	2.3	Virginica

PROGRAM

```
import pandas as pd
from sklearn.neighbors import KNeighborsClassifier
df = pd.read_csv("iris3.csv")
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
model = KNeighborsClassifier(n_neighbors=3)
model.fit(X, y)
sample = [[5.9, 2.8, 4.7, 1.2]]
print("Predicted Flower:", model.predict(sample)[0])
```

OUTPUT

Predicted Flower: Versicolor

d) Iris Dataset 4

Sepal Length	Sepal Width	Petal Length	Petal Width	Species
5.1	3.8	1.5	0.3	Setosa
4.7	3.2	1.3	0.2	Setosa
6.2	2.9	4.3	1.3	Versicolor
5.6	2.9	3.6	1.3	Versicolor
6.8	3.2	5.9	2.3	Virginica
7.1	3.0	5.9	2.1	Virginica

PROGRAM

```
import pandas as pd

from sklearn.neighbors import KNeighborsClassifierdf =
pd.read_csv("iris4.csv")

X = df.iloc[:, :-1]
y = df.iloc[:, -1]

model = KNeighborsClassifier(n_neighbors=3)
model.fit(X, y)

sample = [[6.9, 3.1, 5.8, 2.2]]

print("Predicted Flower:", model.predict(sample)[0])
```

OUTPUT

Predicted Flower: Virginica

e) Iris Dataset 5

Sepal Length	Sepal Width	Petal Length	Petal Width	Species
5.0	3.5	1.3	0.3	Setosa
5.3	3.7	1.5	0.2	Setosa
5.9	3.0	4.2	1.5	Versicolor
6.0	2.9	4.5	1.5	Versicolor
6.5	3.2	5.1	2.0	Virginica
6.7	3.3	5.7	2.1	Virginica

PROGRAM

```
import pandas as pd
from sklearn.neighbors import KNeighborsClassifier
df = pd.read_csv("iris5.csv")
X = df.iloc[:, :-1]
y = df.iloc[:, -1]
model = KNeighborsClassifier(n_neighbors=3)
model.fit(X, y)
sample = [[5.8, 3.0, 4.1, 1.4]]
print("Predicted Flower:", model.predict(sample)[0])
```

OUTPUT

Predicted Flower: Versicolor